

Title: RGB–Depth Fusion with Vision Foundation Models for Semantic Segmentation

Author: Francisco Ribeiro – up201809215@edu.fe.up.pt

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Project Description

Fusing RGB and Depth information remains a central challenge in multimodal visual perception. While early and mid-level fusion strategies are traditionally used, the emergence of vision foundation models (VFMs) enables new directions for multimodal learning. This project investigates RGB-Depth fusion by leveraging pretrained vision models, particularly DINOv2, to build and evaluate fusion strategies that enhance semantic segmentation performance.

The focus has shifted from contrastive learning to the design and analysis of multiple RGB-D fusion architectures that exploit the representational power of VFMs. Rather than learning representations from scratch, this work integrates RGB and Depth features extracted from DINOv2 backbones through carefully designed fusion modules. The objective is to assess how different fusion approaches affect the performance of downstream semantic segmentation tasks.

As a benchmark, the project compares its methods with the state-of-the-art depth-augmented U-Net introduced in "Fine-tuned depth-augmented U-Net for enhanced semantic segmentation in indoor autonomous vision systems." This method, based on an enhanced U-Net architecture, achieves strong results on the NYUv2 dataset. By contrasting the performance of modern VFM-based fusion strategies with this traditional yet highly optimized U-Net variant, the project aims to reveal the trade-offs between architectural simplicity and the potential of large-scale pretrained representations.

This work contributes to the growing field of multimodal foundation model adaptation and evaluates the extent to which depth can be effectively utilized when fused with RGB in a VFM context for dense prediction tasks.

Data Set

This project will use the NYU Depth v2 (NYUv2) dataset [1], a widely used benchmark in RGB-D computer vision. Collected using a Microsoft Kinect sensor, NYUv2 contains over 1,400 labeled indoor scene frames captured across a variety of environments such as offices, bedrooms, and kitchens. Each frame includes a high-resolution RGB image, a spatially aligned depth map, a label for the scene type of the capture and dense semantic segmentation labels for over 800 object categories. Its moderate size and rich annotations make it well-suited for prototyping contrastive representation learning and evaluating aligned embeddings on downstream tasks such as scene classification and segmentation.

References

- [1] N. Silberman, D. Hoiem, P. Kohli, and R. Fergus, "Indoor segmentation and support inference from RGBD images," in *Proceedings of the European Conference on Computer Vision (ECCV)*, pp. 746–760, Springer, 2012.
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